Project report MODSIM DD1354

Simulation of Tennis Ball Trajectories

Per Sjöstedt

persjost@kth.se

Albin Speijer,

speijer@kth.se

Abstract

1. Introduction

1.1 Background

The motion of a tennis ball is influenced by several physical factors such as gravity, air resistance and spin. When a tennis ball is hit, the rotation of the ball generates aerodynamic forces that affect its trajectory through the Magnus effect. This effect produces a lift force perpendicular to the direction of motion, causing the ball to curve in the air depending on the type of spin applied [1], [2].

In addition to this lift force, aerodynamic drag acts opposite to the direction of motion and gradually slows the ball down during flight. As a result, different types of spin, such as top-spin, under-spin and side-spin, produce noticeably different flight paths. These aerodynamic effects play an important role in determining how the ball moves through the air and how quickly it reaches the court surface [1], [2]. In addition to the motion in the air, the interaction between the ball and the court during a bounce also plays a major role in the overall trajectory of the strike.

The bounce behaviour depends on properties such as the coefficient of restitution (COR), which describes how much velocity is retained after the impact, and the coefficient of friction (COF) between the ball and the court surface. These factors determine how both the velocity and the spin of the ball change during the bounce [3], [4]. Understanding how these forces affect the motion of a tennis ball is important in sports physics and helps explain how spin and court surface properties influence the behaviour of the ball during play. Numerical simulations provide a useful way to analyse these effects without the need for complex experimental setups.

1.2 Purpose

The purpose of this project is to simulate the motion and bounce behaviour of a tennis ball using simplified physical models and numerical methods. The simulation models the forces acting on the ball during flight, including gravity, aerodynamic drag and lift generated by spin (Magnus effect), as well as the interaction between the ball and the court during the bounce.

By combining these physical models, the simulation aims to reproduce realistic tennis ball trajectories and bounce dynamics.

1.3 Research Question

The main question of this project is whether a simplified physical model can produce realistic tennis ball trajectories and bounce behaviour. By comparing the simulation results with experimentally

observed behaviour of tennis balls reported in the literature, we evaluate whether the model produces physically reasonable results.

To investigate the influence of spin on the trajectory, all physical parameters in the simulation are kept constant except for the direction of the ball's spin. The simulation compares four cases with the same spin magnitude (3000 rpm): top-spin, under-spin, and side-spin (left and right). This allows us to analyse how the direction of spin affects both the flight trajectory of the ball and its behaviour during the bounce.

2. Methodology

2.1 Flight Physics

During the flight of the tennis ball we can assume that only three forces are acting on it: gravitation, aerodynamic drag and the lift force as a result of the Magnus Effect [2].

2.1.1 Gravitation

To simulate the gravitational force on the ball, we used a version of Newton's universal law of gravitation approximated to an object on the surface of the earth:

$$\vec{F} = m \cdot \vec{g} \tag{1}$$

where F is the gravitational force, m is the mass of an object and g is the gravitational acceleration.

2.1.2 Drag

The aerodynamic drag force acting opposite to the tennis ball's velocity can be calculated given this equation [2]:

$$\vec{F}_D = -\frac{1}{2}\rho A C_D V \cdot \vec{V} \tag{2}$$

where $\vec{F}_D$ is the drag force, ρ is the air density, A is the cross sectional area of the ball, C_D is the dimensionless drag coefficient, V is the magnitude of the velocity and $\vec{V}$ is the velocity vector.

2.1.3 Lift (Magnus Effect)

The Magnus effect can be described as a lift force acting on the tennis balls mass depending on the velocity and spin or angular velocity [2]. This force is perpendicular to the velocity and can be calculated with this equation:

$$\vec{F}_L = \frac{1}{2}\rho A C_L V \cdot \left(\frac{\vec{\omega} \times \vec{V}}{\omega}\right) \tag{3}$$

where $\vec{F}_L$ is the lift force, ρ is the air density, A is the cross sectional area of the ball, C_L is the dimensionless lift coefficient, V is the magnitude of the velocity, $\vec{\omega}$ is the angular velocity vector, $\vec{V}$ is the velocity vector and ω is the magnitude of the angular velocity.

2.1.4 Drag and Lift Coefficients

The drag and lift coefficients depend on the velocity and angular velocity of the tennis ball. According to empirical measurements, linear relations for these coefficients have been derived [2]:

$$C_D = 0.6204 - 9.76 \cdot 10^{-4} \cdot (V - 50) + [1.027 \cdot 10^{-4} - 2.24 \cdot 10^{-6} \cdot (V - 50)]\omega \tag{4}$$

$$C_L = [4.68 \cdot 10^{-4} - 2.0984 \cdot 10^{-5} \cdot (V - 50)]\omega \tag{5}$$

where V is the magnitude of the velocity and ω is the magnitude of the angular velocity.

2.1.5 Moment of inertia

The moment of inertia of the tennis ball is approximated as that of a hollow spherical shell, since most of the mass is located in the outer rubber shell [3] . The moment of inertia is therefore defined as

$$I = \frac{2}{3} m r^2 \tag{6}$$

where m is the mass of the ball and r is the radius of the ball.

2.2 Bounce Physics

During contact of the tennis ball and the underlying court, we can assume that three forces act on the ball: gravitation, normal force and a friction force. This model is taken from Rod Cross experiments of a tennis ball bouncing against a tennis racket [5]. Our case has a tennis ball bouncing against a tennis court but we are using the same model.

According to this model, the friction can be divided into two cases depending on the spin of the ball during contact. Either the ball is sliding against the court, which causes regular sliding friction, or the ball rolls against the court (gripping).

2.2.1 Gravitation

The gravitational force is calculated in the exact same way as in equation 1.

2.2.2 Normal Force

We approximated the bounce of the ball against the court as a mass-spring-damper system that goes through half a period. In such a system the force can be described as a differential equation [6, Ch. 2] that can be rewritten for our case as:

$$\vec{N} = k \cdot \vec{\Delta y} - c \cdot \vec{v}_y \tag{7}$$

where $\vec{N}$ is the force acting on the object, in our case the normal force, k is the spring constant, $\vec{\Delta y}$ is the vertical displacement, c is the damping constant and $\vec{v}_y$ is the vertical velocity.

As further developed in 3.1 the damping constant can be calculated from the damping ratio. This damping ratio is being calculated from the coefficient of restitution. The formulas being used are the following:

$$\zeta = \frac{-\ln COR}{\sqrt{\pi^2 + (\ln COR)^2}} \tag{8}$$

$$c = 2 \cdot \zeta \cdot \sqrt{km} \tag{9}$$

where ζ is the damping ratio, COR is the coefficient of restitution, c is the damping contant, k is the spring constant and m is the mass of the tennis ball.

2.2.3 Friction

According to Cross [5], the condition for when gripping occurs is:

$$v_{rel} = v_{x,z} - r \cdot \omega = 0 \tag{10}$$

where v_{rel} is the relative velocity of the ball's surface, $v_{x,z}$ is the horizontal components of the ball's velocity, r is the ball's radius and ω is the angular velocity.

In our case we need the relative velocity as a vector for deciding the direction of the friction force since these have opposite directions. To get the relative velocity as a vector we modified this equation to:

$$\vec{v}_{rel} = \vec{v}_{x,z} - \vec{r} \times \vec{\omega} \tag{11}$$

to see if:

$$|\vec{v}_{rel}| = 0 \tag{12}$$

If the condition in equation (12) is not fulfilled, sliding friction will act on the ball. The sliding friction force has the opposite direction to the relative velocity and can be calculated from this equation:

$$\vec{F}_{friction} = -\frac{\vec{v}_{rel}}{|\vec{v}_{rel}|} \cdot COF \cdot N \tag{13}$$

where COF is the coefficient of friction and N is the magnitude of the normal force.

If the condition in equation (12) is fulfilled or close to zero, the ball rolls against the court and deforms elastically [5]. We approximated this as a regular mass spring system that deforms according to this formula:

$$\vec{F}_{friction} = -k \cdot \Delta\vec{s} \tag{14}$$

where k is the spring constant, and $\Delta\vec{s}$ is the horizontal distance from the point of impact to the current position of the ball. This distance we attained by performing the Euler method on the relative velocity:

$$\Delta\vec{s}_{n+1} = \Delta\vec{s}_{n} + \vec{v}_{rel} \cdot dt \tag{15}$$

where $\Delta\vec{s}_{n+1}$ is the horizontal distance in the next time step, $\Delta\vec{s}_{n}$ is the current distance, $\vec{v}_{rel}$ is the relative velocity and dt is the time step.

Something worth pointing out is that we call the elastic force on the ball while gripping $\vec{F}_{friction}$ even though it is not a friction force, that was simply a naming convention to simplify our code. The simulation was also limited so that if this elastic force reached the same magnitude as the sliding friction, the ball was simulated to be sliding instead.

2.2.4 Torque

The torque on the ball during contact with the ground can be calculated using the vector torque equation:

$$\vec{\tau} = \vec{r} \times \vec{F} \tag{16}$$

where $\vec{\tau}$ is the torque, $\vec{r}$ is the radius as a vector from the point that the torque is measured at (the center of the ball) to the point that the force acts on, $\times$ symbolises a cross product and $\vec{F}$ is the force that creates the torque.

2.3 Numerical Integration

To simulate the trajectory of the tennis ball in Unity we decided to use numerical integration to calculate the position and angular velocity of the ball based on the forces acting on it.

2.3.1 Position

The simulation is heavily based on the differential equation that can be derived from Newton's second law:

$$\vec{F} = m \cdot \vec{a} \tag{17}$$

where $\vec{F}$ is the net force, m is the mass of the object and $\vec{a}$ is the acceleration.

The acceleration of an object is the time derivative of the velocity and the velocity of an object is the time derivative of the position. Therefore we can attain the velocity by integrating the acceleration and the position by integrating the velocity. For this integration we used the semi-implicit Euler Method accordingly:

$$\vec{v}_{n+1} = \vec{v}_n + \vec{a}_n \cdot dt \tag{18}$$

where $\vec{v}_{n+1}$ is the velocity in the next time step, $\vec{v}_n$ is the current velocity, $\vec{a}_n$ is the current acceleration and dt is the time step.

$$\vec{pos}_{n+1} = \vec{pos}_n + \vec{v}_{n+1} \cdot dt \tag{19}$$

where $\vec{pos}_{n+1}$ is the position in the next time step, $\vec{pos}_n$ is the current position, $\vec{v}_{n+1}$ is the newly calculated velocity in the next time step and dt is the time step.

2.3.2 Angular Velocity

The simulation for the angular velocity was done similarly. This differential equation can be derived from Newton's second law of rotation:

$$\vec{\tau} = I \cdot \vec{\alpha} \tag{20}$$

where $\vec{\tau}$ is the torque, I is the moment of inertia and $\vec{\alpha}$ is the angular acceleration.

Since the angular velocity is the time derivative of the angular acceleration we used Euler's method similarly as in 2.3.1:

$$\vec{\omega}_{n+1} = \vec{\omega}_n + \vec{\alpha}_n \cdot dt \tag{21}$$

where $\vec{\omega}_{n+1}$ is the angular velocity in one moment, $\vec{\omega}_n$ is the angular velocity in the previous moment, $\vec{\alpha}_n$ is the angular acceleration in the previous moment and dt is the time step.

2.4 Comparison of Simulation

To measure how well our simulation compares to real tennis ball trajectories we used initial positions, velocities and launch angles as values in simulations done by Rod Cross [1] and Gary and Ian Robinson [2]. To compare results we measured the position of the first impact against the court.

To evaluate how well our approximation of the mass-spring-damper model simulates the normal force during contact with the ground we measured the peak height of the ball between the first and the second bounce against the ground. This was performed by simulating dropping a ball from 1m with no initial velocity. According to Brody [3] a tennis ball dropped from 100 inches (2.54m) must have a return height of 53-58 inches (1.3462-1.4732m) in order for a tennis ball to be approved. This is what we will compare our result to.

3. Implementation

3.1 Parameters

The simulation uses a set of physical parameters to describe the tennis ball, the aerodynamic environment, and the interaction between the ball and the court surface. The mass and radius of the tennis ball as well as court and net dimensions are taken from standard International Tennis Federation (ITF) specifications [7]. The coefficients of restitution and friction for a tennis ball against a court is taken as a value in the spans given by Rod Cross [8]. Cross also mentions in another report [4] an approximate value for the spring constant of a tennis ball. Table 1 summarizes the constants used throughout the simulation.

Table 1: Physical parameters and constants used in the simulation.

Parameter	**Symbol**	**Value**	**Unit**
Radius (tennisball)	r	0.033	m
Weight (tennisball)	m	0.058	kg
Court Dimensions	-	10.97x23.77	m^2
Net Height (middle)	-	0.914	m
Gravitational acceleration	g	9.81	m/s^2
Air density (20C)	ρ	1.22	kg/m^3
Spring constant	k	20 000	N/m
Coefficent of restitution	COR	0.75	–
Coefficient of friction	COF	0.7	–

The cross-sectional area of the ball, which represents the projected area moving through the air, is calculated from the radius as

$$A = \pi r^2 \tag{22}$$

To simulate the deformation of the ball during impact, a spring-dampener model is used. The spring constant produces a realistic compression of the ball during contact.

The damping constant c is calculated from the damping ratio, ζ, which depends on the logarithmic decrement, δ, which is defined as the ratio of amplitudes of a damped oscillation [6, Ch. 2]. In our case we calculated the logarithmic decrement from the constant of restitution, since that comes from a ratio of the difference in height before and after a bounce. By linking the coefficient of restitution to the logarithmic decrement we can calculate the damping ratio and therefore the damping constant from the coefficient of restitution. The equations used for these calculations are (8) and (9).

3.2 Program structure

The code mainly consists of a loop that runs on every frame that runs on a fixed time interval called FixedUpdate, which is built-in to Unity. We set this time interval to 0.0005s to make sure that the simulation will be finer in detail while still running smoothly on our computer.

For keeping track of when the ball first comes in contact with the ground during a bounce, a boolean variable, wasInAir, is set to either true or false. If the vertical position is at or below the contact point with the ground the variable and wasInAir is set to true, we add to a variable, bounceCounter, that the ball has done another bounce.

Every iteration of this loop the totalForce-variable is set to the gravitational force. After that, depending on if the vertical position is above or below the contact point with the ground, which is y = 0 + radius, we add the forces from our methodology to that variable. We made a method for each case to apply the appropriate forces. Additionally in the method for the bounce forces, we handle the numerical integration for calculating the angular velocity.

After applying the forces to the totalForce variable, we handle the numerical integration for calculating the position of the ball. Since we now have the position and angular velocity we can update the Unity-gameObject to act according to those numbers.

Finally, the simulation is stopped if one of two conditions are fulfilled. The first is if the bounceCounter variable reaches two, since the hypothetical opponent in a tennis match would have to respond before that. The second is if the ball “hits the net”. This happens if the x-coordinate of the position of the ball is close to zero ($\pm$1cm) and the y-coordinate is at or below the net height.

4. Results

4.1 Validation of simulation

In Cross’s work [1], two balls were being hit at the baseline, 11.885m behind the net, and a height of 1m. One with a top-spin of 20 revolutions per second or 1200 RPM, the other one had no spin. He created two scenarios: one where the balls just cleared the net and one where the balls first impact against the court was on the opposing baseline. The balls had an initial velocity of 30 m/s. Impact position is measured by how far away from the net the ball first touches the court.

Table 2: Impact positions for Cross compared to our simulation.

Spin Type	Elevation Angle [°]	Impact Position [m] (Cross)	Impact Position [m] (our simulation)
Top-spin	11.9	11.885	11.381
No spin	8.1	11.885	10.767

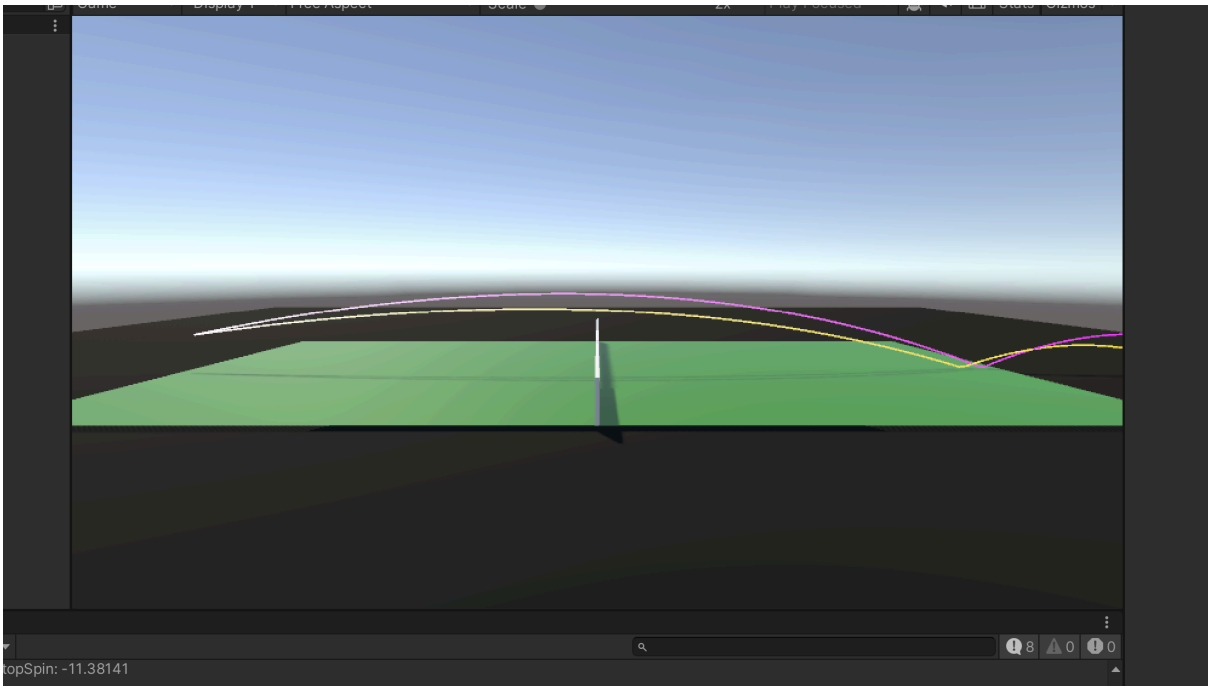

Figure 1: Trajectories for top-spin (purple) and no spin (yellow).

For the other comparison the trajectory of a serve is modeled [2] based on Robinsons and Robinsons work. The simulation constants used here is for a top-spin serve (also known as kick serve). The initial velocity set to 44.44 m/s, height from baseline of 3 meters and an elevation angle of -1.55°.

Table 3: Impact positions for Robinson & Robinson compared to our simulation.

Spin magnitude	Impact position [m] (Robinson & Robinson)	Impact position [m] (our simulation)
None	12.825	12.845
1909 RPM (200 rad/s)	8.055	7.997
3819 RPM (400 rad/s)	5.225	5.169

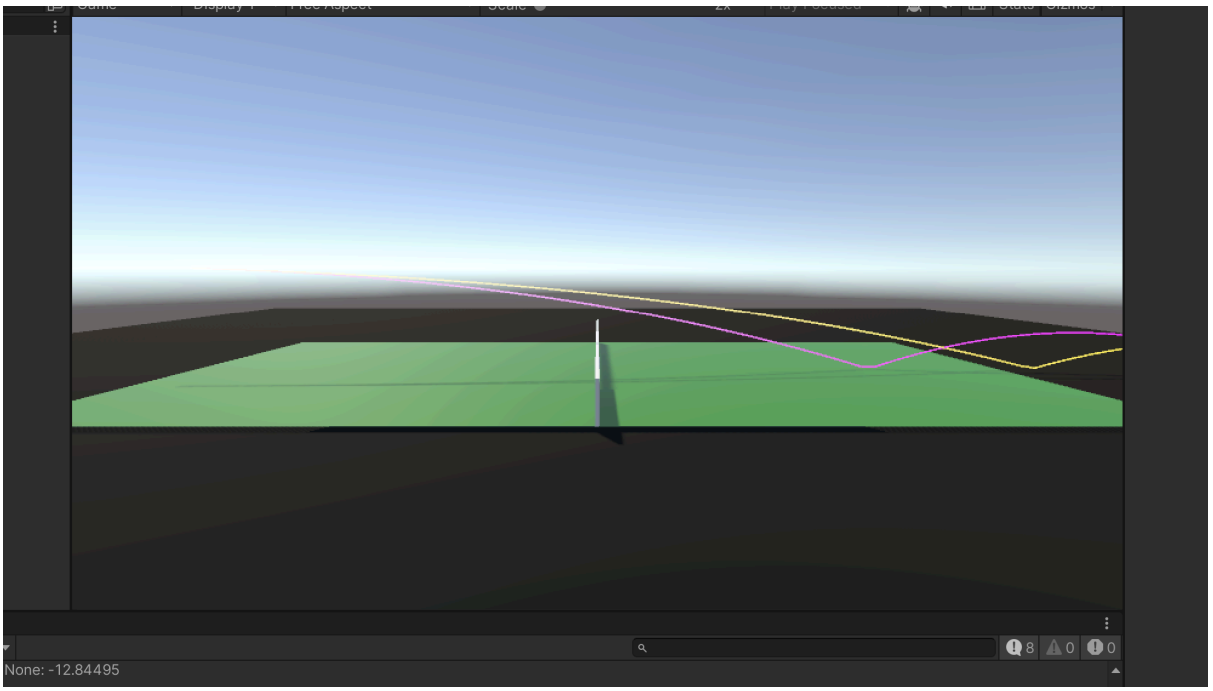

Figure 2: Trajectories for 200 rad/s top-spin serve (purple) and no spin serve (yellow).

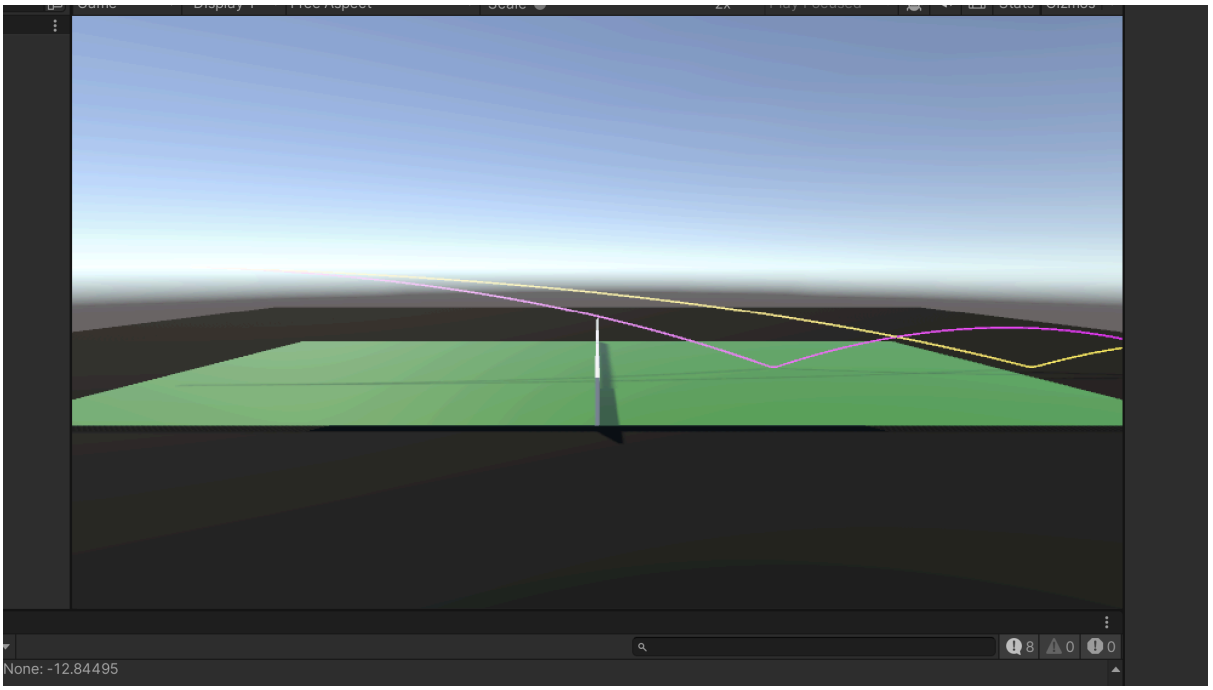

Figure 3: Trajectories for 400 rad/s top-spin serve (purple) and no spin serve (yellow).

4.2 Simulation for Different Spin Types

The ball is struck from 11.885m behind the net and from a height of 1m. The ball is also struck from the middle of the court's width (z = 0).

The initial simulation constants were set to:
Initial velocity = 35m/s
Elevation angle = 7°
Spin = 3000RPM

Table 4: Impact positions and velocities for different types of spin.

Spintype	**First bounce [m] (x)**	**First bounce [m] (z)**	**Second bounce [m] (x)**	**Second Bounce [m] (z)**	**Second bounce [m/s] (velocity)**
Top	5.20	0	14.35	0	14.266
Under	28.53	0	33.06	0	6.38
Left	12.88	3.25	19.78	5.61	10.5445
Right	12.88	-3.25	19.78	-5.61	10.5445

The x-positions show how far past the net the ball has travelled. As mentioned earlier, the court is 23.77m which means that half of the court is 11.885m. Positive z-values show how much to the right the ball has travelled from the center line and vice versa.

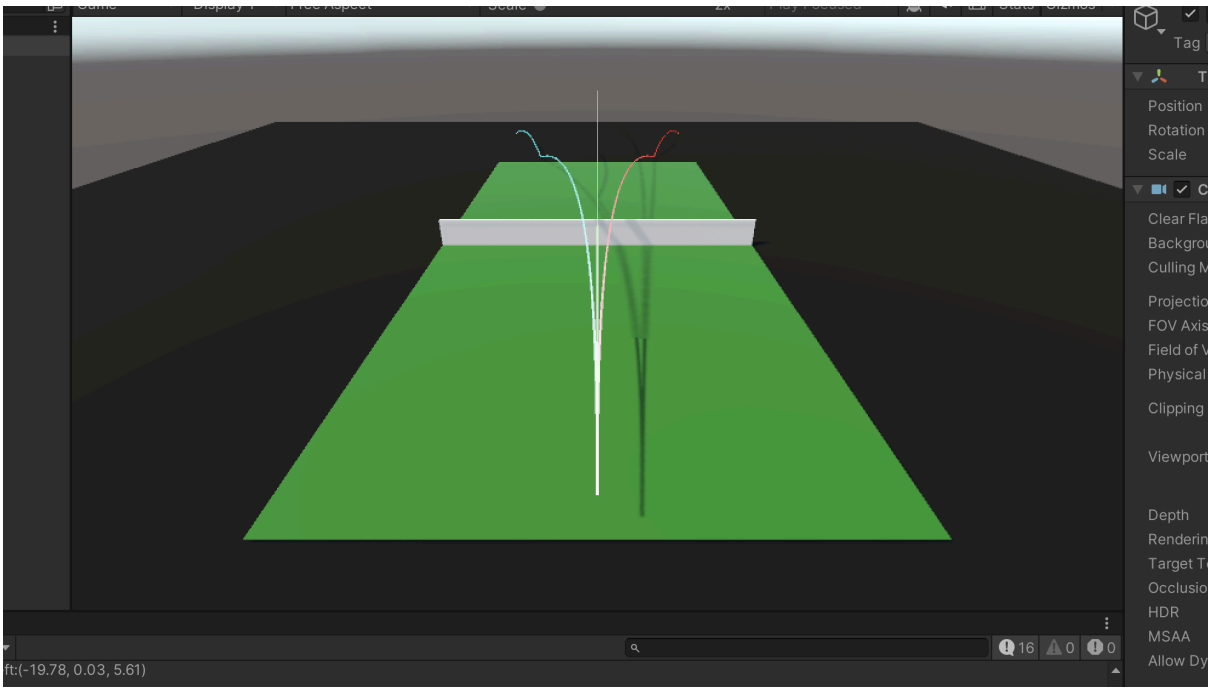

Figure 4: Trajectories of different spins (green = under-spin, red = side-spin left, blue = side-spin right, purple = top-spin)

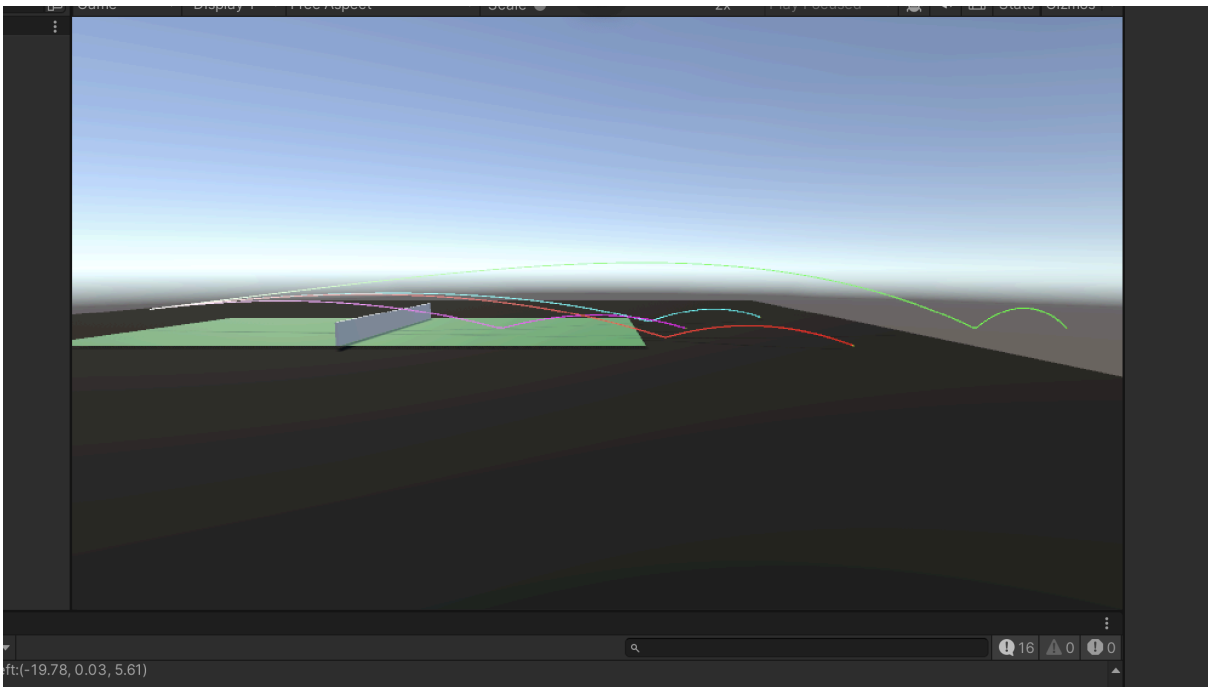

Figure 5: Trajectories of different spins (green = under-spin, red = side-spin left, blue = side-spin right, purple = top-spin)

Return height, or peak height between the first and second bounce, of dropping a ball from 2.54m measured at 1.33m in our simulation of this scenario.

5. Discussion

A comparison of our simulation to earlier work is essential to verify the realism of our model. If we begin to compare the results of Cross's simulation [1] we can see that the difference in distance traveled is 0.504 meters for no spin and 1.118 meters for the top-spin with 20 rev/s (or 1200RPM). This is a significant difference that could depend on many factors.

For the comparison against Robinson and Robinson's work [2] the differences are negligible. For the different variants of serve (top-spin or kick serves) the differences are 0.02m, 0.058m and 0.056m (for no spin, 200 rad/s and 400 rad/s respectively). Small differences are to be expected since any minor factor could affect the end result. For example choice of different COR, air density for example, or simply a different physics implementation. That the results end up this similar can therefore to some extent confirm our model.

The differences can with high likelihood be attributed to the calculation of C_L and C_D where Cross's simulation of aerodynamic forces treats the drag coefficient as mostly constant for typical ball speeds. In comparison, our model does the same implementation as Robinson and Robinson's. These values are taken from linear interpolation around observed data (50 m/s). This dynamically updates the values of the coefficients based on current angular velocity and current speed.

For our simulation on its own the different types of spin are compared when the other parameters are kept constant. For a spin of 3000RPM (see table 4) the results are drastically different. Most noticeable is the difference between under-spin and top-spin. For this setup a difference of 23.33 meters can be observed for the first bounce and a difference of 18.71 meters on the second bounce, as well as a difference of around 8m/s in speed. This can be attributed to the Magnus effect. That a high RPM top-spin dives is exactly what is to be expected and why professional tennis players try to achieve a top-spin. That the ball dives quickly means you can strike at a high speed and still hit the ground before the opposing baseline. On the contrary, under-spins as we can see must be hit at a much lower speed and elevation angle to be inside of the court.

By comparing Brody's return height after one bounce from being dropped to ours we find that our return height is in the range that Brody mentions makes a ball approved. This can be used to evaluate how well the mass-spring-damper model approximated the normal force during contact with the ground. Since the return height falls in the range of approval it seems to be a well functioning approximation for this drop simulation. Brody mentions that the coefficient of restitution differs depending on the drop height so the fact that we use a constant value, based on a drop from 2.54 meters, makes the approximation less realistic the further our simulation strays away from that drop. This means that this probably has an impact on our simulations of a strike from a height of 1m or the serve from a height of 3m.

The choice of numerical method used in this simulation is semi-implicit Euler. This was used since explicit Euler tends to add energy in harmonic oscillators [9], which we simulate during the bounce. The choice for using semi implicit instead of strictly implicit was to get higher accuracy of the position calculation by using the newly calculated velocity instead of using that of the previous time step.

Limitations / improvements
While the model demonstrates strong alignment with previous work some limitations must be acknowledged. One example is that our coefficients of lift and drag are optimized for high speed strikes at around 50 m/s. At much lower speeds these interpolations may therefore lose accuracy. To compensate for this a lower bound was applied. For the spring constant the model assumes a constant spring stiffness (k) and treats the ball as a perfect sphere. In reality the tennis ball would deform and have a different cross sectional area. In the case of a tennis ball bouncing on a racket [5], Cross explains that the sliding and gripping forces are more complex than the model we used as they combine and can switch directions. Future improvements could include adding spin decay over time as well improved calculations of sliding and gripping.

7. Workflow

The workload for this project was distributed very evenly. Most things were done in equal load. The programming was mostly done together while discussing. If anything the heavier coding work was done more by Per Sjöstedt and more of the source gathering and report was done by Albin Speijer but all aspects of the project was done under close collaboration. It has been helpful that we have worked together on similar projects before. The project has gone well but if we were to do something different the work on the report would have been started earlier so that we had more time for the discussion part of the project.

8. References

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